

Worksheet 5: Terrestrial Planets

Full solutions

Covers Lectures 9 and 10: equilibrium temperature, the runaway greenhouse, deuterium and water loss, Jeans escape, Mercury and Mars

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that..."); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end. All parts are analytical; no computer is needed.

Constants and data

$\sigma = 5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$, $k_B = 1.381 \times 10^{-23} \text{ J K}^{-1}$, $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, $m_u = 1.661 \times 10^{-27} \text{ kg}$, $S_\oplus = 1361 \text{ W m}^{-2}$ (at 1 AU)

Water and grey atmosphere: latent heat $L = 2.5 \times 10^6 \text{ J kg}^{-1}$, vapour gas constant $R_v = 461 \text{ J kg}^{-1} \text{ K}^{-1}$, reference $(p_0, T_0) = (611 \text{ Pa}, 273 \text{ K})$, grey opacity $\kappa = 5 \times 10^{-2} \text{ m}^2 \text{ kg}^{-1}$, $g = 10 \text{ m s}^{-2}$

Earth: $A = 0.30$, $T_{\text{surf}} = 288 \text{ K}$, $a = 1.00 \text{ AU}$. Venus: $A = 0.77$, $a = 0.723 \text{ AU}$, $S = 1.91 S_\oplus$, $T_{\text{surf}} = 737 \text{ K}$

Mars: $M = 6.4 \times 10^{23} \text{ kg}$, exobase radius $r_{\text{exo}} = 3.6 \times 10^6 \text{ m}$, exobase temperature $T_{\text{exo}} = 270 \text{ K}$, rotation period 24.62 h

Mercury: $R = 2440 \text{ km}$, spin period $P_{\text{rot}} = 58.65 \text{ d}$, orbital period $P_{\text{orb}} = 87.97 \text{ d}$, radius change $\Delta R \approx 7 \text{ km}$.

Relative molecular masses (in units of m_u): $\text{H} = 1$, $\text{H}_2 = 2$, $\text{CO}_2 = 44$

Atmospheric and escape data as in the Lecture 9 and 10 notes.

Problem 1 Equilibrium temperature and the greenhouse effect

The equilibrium temperature is the first number to compute for any planet: it is what the surface would reach with no atmosphere, set by the balance of absorbed sunlight against emitted infrared. The gap between it and the true surface temperature measures the greenhouse effect.

(a) The energy balance for a rapidly rotating planet of Bond albedo A at insolation S is $\pi R^2 S(1 - A) = 4\pi R^2 \sigma T_{\text{eq}}^4$, so $T_{\text{eq}} = [S(1 - A)/4\sigma]^{1/4}$. Compute Earth's equilibrium temperature.

Solution.

$$T_{\text{eq}} = \left[\frac{S_{\odot}(1-A)}{4\sigma} \right]^{1/4} = \left[\frac{1361 \times 0.70}{4 \times 5.670 \times 10^{-8}} \right]^{1/4} = \left[\frac{952.70}{2.2680 \times 10^{-7}} \right]^{1/4},$$

which gives

$$T_{\text{eq}} = (4.2006 \times 10^9)^{1/4} = 254.58 \text{ K} \approx 255 \text{ K}.$$

The factor of 4 is the ratio of the sphere's surface area to its intercepting disc: the planet catches sunlight on a disc of area πR^2 and radiates from the whole sphere of area $4\pi R^2$.

| $T_{\text{eq}} = 254.58 \text{ K} \approx 255 \text{ K}$; **the factor of 4 accounts for intercepting solar flux on area πR^2 while radiating over the entire sphere $4\pi R^2$.**

(b) A single infrared-opaque layer in radiative equilibrium re-radiates downward as well as upward, so the surface energy balance becomes $\sigma T_s^4 = 2\sigma T_{\text{eq}}^4$. Show that $T_s = 2^{1/4} T_{\text{eq}}$, evaluate it for Earth, and compare with the observed surface temperature of 288 K.

Solution. The opaque layer absorbs the surface infrared and re-emits σT_a^4 both up and down. Radiative equilibrium at the top of the atmosphere requires $T_a = T_{\text{eq}}$, and the surface then receives the absorbed sunlight plus the downward emission, $2\sigma T_{\text{eq}}^4$, giving

$$T_s = 2^{1/4} T_{\text{eq}} = 1.18921 \times 254.58 = 302.75 \text{ K}.$$

The one-layer model over-predicts the observed 288 K by about 15 K. A single fully opaque layer is too strong a greenhouse for Earth, whose atmosphere is only partly opaque in the infrared. The model captures the mechanism, downward re-radiation warming the surface above T_{eq} , but not the magnitude.

| $T_s = 302.75 \text{ K}$, **which over-predicts the observed 288 K by about 15 K because Earth's atmosphere is only partly opaque in the infrared.**

(c) Compute Venus's equilibrium temperature from its own albedo and insolation, compare it with Earth's, and state the greenhouse warming $\Delta T = T_{\text{surf}} - T_{\text{eq}}$ for each planet.

Solution.

$$T_{\text{eq}}^{\text{Venus}} = \left[\frac{1.91 \times 1361 \times (1 - 0.77)}{4 \times 5.670 \times 10^{-8}} \right]^{1/4} = \left[\frac{2599.5 \times 0.23}{2.2680 \times 10^{-7}} \right]^{1/4},$$

so

$$T_{\text{eq}}^{\text{Venus}} = (2.6362 \times 10^9)^{1/4} = 226.59 \text{ K} \approx 227 \text{ K}.$$

Venus's equilibrium temperature is lower than Earth's (227 against 255 K), even though Venus is closer to the Sun, because its bright clouds reflect 77% of the incident sunlight. The greenhouse warmings are

$$\Delta T_{\text{Earth}} = 288 - 254.58 = 33.42 \text{ K}, \quad \Delta T_{\text{Venus}} = 737 - 226.59 = 510.41 \text{ K}.$$

The two planets have similar equilibrium temperatures but greenhouse effects that differ by a factor of fifteen.

| $T_{\text{eq}}^{\text{Venus}} = 226.59 \text{ K} \approx 227 \text{ K}$ (**below Earth's 255 K**); **greenhouse warmings are $\Delta T_{\text{Earth}} = 33.42 \text{ K}$ and $\Delta T_{\text{Venus}} = 510.41 \text{ K}$, differing by a factor of fifteen.**

Problem 2 The runaway greenhouse

A planet stays habitable only while its atmosphere can radiate away the sunlight it absorbs. A moist atmosphere has a ceiling on how much infrared it can emit, and crossing it triggers a runaway greenhouse. This problem locates that ceiling and applies it to early Venus.

(a) Consider a young terrestrial planet with liquid water and Earth-like water clouds, so $A = 0.30$. Compute the absorbed solar flux per unit surface area, $F_{\text{abs}} = S(1 - A)/4$, for a planet at Earth's orbit and for one at Venus's orbit.

Solution.

$$F_{\text{abs}}^{\text{Earth}} = \frac{1361 \times 0.70}{4} = 238.18 \text{ W m}^{-2}, \quad F_{\text{abs}}^{\text{Venus}} = \frac{1.91 \times 1361 \times 0.70}{4} = 454.91 \text{ W m}^{-2}.$$

At a common albedo the only difference is the insolation: Venus's orbit delivers 1.91 times Earth's, so it absorbs 1.91 times as much. This is the flux the atmosphere must radiate back to space to hold a steady temperature.

$F_{\text{abs}}^{\text{Earth}} = 238.18 \text{ W m}^{-2}$ and $F_{\text{abs}}^{\text{Venus}} = 454.91 \text{ W m}^{-2}$; at equal albedo Venus absorbs 1.91 times more flux purely due to its higher insolation.

(b) In a steam atmosphere the infrared photosphere forms where the overlying water-vapour column reaches optical depth of order unity, at pressure $p_{\text{phot}} = g/\kappa$. Keep the planet at Earth's orbit from part (a), and take g and κ from the Constants and data box. Compute p_{phot} , then use the integrated Clausius-Clapeyron relation

$$\ln \frac{p}{p_0} = -\frac{L}{R_v} \left(\frac{1}{T} - \frac{1}{T_0} \right)$$

to find the temperature at which saturated water vapour reaches this pressure.

Solution.

$$p_{\text{phot}} = \frac{g}{\kappa} = \frac{10}{5 \times 10^{-2}} = 200 \text{ Pa}.$$

Invert Clausius-Clapeyron for T at $p = 200 \text{ Pa}$, with $L/R_v = 2.5 \times 10^6 / 461 = 5423.0 \text{ K}$:

$$\frac{1}{T_{\text{phot}}} = \frac{1}{T_0} - \frac{R_v}{L} \ln \frac{p_{\text{phot}}}{p_0} = 3.6630 \times 10^{-3} - \frac{\ln(200/611)}{5423.0},$$

which evaluates to

$$\frac{1}{T_{\text{phot}}} = 3.6630 \times 10^{-3} + 2.0593 \times 10^{-4} = 3.8689 \times 10^{-3} \text{ K}^{-1},$$

so $T_{\text{phot}} = 258.47 \text{ K} \approx 258 \text{ K}$. The photosphere temperature is fixed by the vapour-pressure curve alone: it does not depend on the surface temperature below it. This is the key to the runaway.

$p_{\text{phot}} = 200 \text{ Pa}$ and $T_{\text{phot}} = 258.47 \text{ K} \approx 258 \text{ K}$; the photosphere temperature is fixed entirely by the vapour-pressure curve and does not depend on the surface temperature below.

(c) The outgoing longwave radiation of the steam atmosphere cannot exceed the black-body emission of its photosphere, $F_{\text{OLR}}^{\text{max}} = \sigma T_{\text{phot}}^4$. Compute this limit, compare it with the

absorbed fluxes from part (a), and state which planet enters a runaway greenhouse.

Solution.

$$F_{\text{OLR}}^{\text{max}} = \sigma T_{\text{phot}}^4 = 5.670 \times 10^{-8} \times (258.47)^4 = 5.670 \times 10^{-8} \times 4.4631 \times 10^9 = 253.06 \text{ W m}^{-2}.$$

The Earth-orbit planet absorbs 238 W m^{-2} , below the ceiling, so it settles into a steady moist climate. The Venus-orbit planet absorbs 455 W m^{-2} , above the ceiling: no surface temperature lets its atmosphere radiate away the absorbed sunlight, so the oceans evaporate entirely and the surface heats without a stable balance until all the water is in the atmosphere. This is the runaway greenhouse. The simple grey estimate places the limit at 253 W m^{-2} ; full radiative-convective models, and the Lecture 9 notes, give a somewhat higher value near 280 W m^{-2} , but the conclusion is unchanged.

$F_{\text{OLR}}^{\text{max}} = 253.06 \text{ W m}^{-2}$: Earth (238 W m^{-2}) stays below the ceiling and remains stable, whereas Venus (455 W m^{-2}) exceeds it and enters a runaway greenhouse.

(d) Explain in words why the outgoing infrared flux of a steam atmosphere saturates at an upper limit, and why this defines the inner edge of the habitable zone.

Solution. As the surface warms, more water evaporates and the atmosphere grows more opaque in the infrared, so the emitting photosphere moves to higher, colder levels. Once the atmosphere is saturated with water vapour, the photosphere pressure is fixed at g/κ and the photosphere temperature is fixed by the vapour-pressure curve, independent of how hot the surface below becomes. The emitted flux σT_{phot}^4 therefore stops rising: it is capped at the Simpson-Nakajima limit. The inner edge of the habitable zone is the orbit where the absorbed solar flux equals this cap. Inside it, absorbed sunlight always exceeds the maximum the atmosphere can radiate, so liquid water cannot persist and any ocean runs away.

Water-vapour opacity fixes the photosphere pressure at g/κ and its temperature via the vapour-pressure curve, capping OLR at the Simpson-Nakajima limit; inside the orbit where absorbed flux exceeds this cap, oceans run away.

Problem 3 Thermal escape from Mars

Whether a planet keeps its atmosphere is decided at the exobase, the level where a molecule makes its last collision and the fastest ones escape. The Jeans parameter compares gravity's grip on a molecule to its thermal energy, and it sorts which gases a planet can hold.

(a) Compute the escape speed at Mars's exobase.

Solution.

$$v_{\text{esc}} = \sqrt{\frac{2GM}{r_{\text{exo}}}} = \sqrt{\frac{2 \times 6.674 \times 10^{-11} \times 6.4 \times 10^{23}}{3.6 \times 10^6}} = \sqrt{\frac{8.5427 \times 10^{13}}{3.6 \times 10^6}},$$

so

$$v_{\text{esc}} = \sqrt{2.3730 \times 10^7} = 4871.3 \text{ m s}^{-1} \approx 4.9 \text{ km s}^{-1}.$$

$v_{\text{esc}} = 4871.3 \text{ m s}^{-1} \approx 4.9 \text{ km s}^{-1}$ at Mars's exobase.

(b) The Jeans escape parameter for a molecule of mass m is $\lambda = GMm/(k_B T_{\text{exo}} r_{\text{exo}})$, the ratio of gravitational to thermal energy at the exobase. Compute λ for atomic hydrogen, molecular hydrogen, and carbon dioxide.

Solution. Compute the mass-independent group first:

$$\frac{GM}{k_B T_{\text{exo}} r_{\text{exo}}} = \frac{6.674 \times 10^{-11} \times 6.4 \times 10^{23}}{1.381 \times 10^{-23} \times 270 \times 3.6 \times 10^6} = \frac{4.2714 \times 10^{13}}{1.3423 \times 10^{-14}} = 3.1820 \times 10^{27} \text{ kg}^{-1}.$$

Multiplying by each molecular mass $m = (\text{relative mass}) \times m_u$:

$$\lambda_{\text{H}} = 3.1820 \times 10^{27} \times 1 \times 1.661 \times 10^{-27} = 5.2854,$$

$$\lambda_{\text{H}_2} = 2 \times 5.2854 = 10.571, \quad \lambda_{\text{CO}_2} = 44 \times 5.2854 = 232.56.$$

The parameter scales linearly with molecular mass: heavier molecules are bound far more tightly.

$\lambda_{\text{H}} = 5.2854$, $\lambda_{\text{H}_2} = 10.571$, and $\lambda_{\text{CO}_2} = 232.56$; the Jeans escape parameter scales linearly with molecular mass, binding heavier species far more tightly.

(c) The Jeans escape flux carries a factor $(1 + \lambda)e^{-\lambda}$ and a thermal-speed factor $\propto m^{-1/2}$. Form the ratio of the escape fluxes of atomic and molecular hydrogen, and state whether CO_2 can escape thermally at all.

Solution. The exponential factors are $e^{-\lambda_{\text{H}}} = e^{-5.2854} = 5.0651 \times 10^{-3}$ and $e^{-\lambda_{\text{H}_2}} = e^{-10.571} = 2.5656 \times 10^{-5}$. The flux ratio is

$$\frac{\Phi_{\text{H}}}{\Phi_{\text{H}_2}} = \sqrt{\frac{m_{\text{H}_2}}{m_{\text{H}}}} \frac{(1 + \lambda_{\text{H}})e^{-\lambda_{\text{H}}}}{(1 + \lambda_{\text{H}_2})e^{-\lambda_{\text{H}_2}}} = \sqrt{2} \times \frac{6.2854 \times 5.0651 \times 10^{-3}}{11.571 \times 2.5656 \times 10^{-5}} = 1.4142 \times 107.25 = 151.67.$$

Atomic hydrogen escapes about 150 times faster than molecular hydrogen, both because it is lighter and, mostly, because its smaller Jeans parameter puts far more of the velocity tail above the escape speed. For carbon dioxide, $e^{-\lambda_{\text{CO}_2}} = e^{-232.56} \approx 10^{-101}$; the thermal escape flux is zero for any practical purpose. Mars cannot lose CO_2 by Jeans escape.

$\Phi_{\text{H}}/\Phi_{\text{H}_2} = 151.67$ (about 150), while for CO_2 the factor $e^{-\lambda_{\text{CO}_2}} \approx 10^{-101}$ means thermal Jeans escape is negligible.

(d) A fellow student writes: “If Mars had kept a stronger magnetic field, it would have retained its CO_2 atmosphere.” Judge this claim.

Solution. A global field slows the loss, so the claim names a real effect, but it overstates what a field does. Jeans escape, the process of parts (b) and (c), is independent of the field, and it cannot remove CO_2 from Mars in any case: the Jeans parameter of CO_2 is so large that thermal escape is negligible. Mars lost its atmosphere through the non-thermal channels: ion pickup and sputtering by the solar wind, and photochemical escape, in which dissociation gives carbon and oxygen atoms enough speed to leave. A global field deflects the solar wind and weakens the two solar-wind channels, so a longer-lived field would have slowed the loss; photochemical escape continues on magnetised and unmagnetised planets alike (Lecture 5). How much a

field protects an atmosphere is debated: MAVEN escape rates show that an unmagnetised Mars loses its atmosphere over billions of years, and Venus holds a 92 bar atmosphere of mostly CO₂ with no detectable field. The defensible version of the claim is that Mars is too small to hold a thick atmosphere once its field is gone, so a longer-lived field would have slowed the loss without settling what Mars keeps.

I Overstated: a global field weakens the solar-wind channels (ion pickup, sputtering) and would have slowed the loss, but it leaves photochemical escape untouched, Jeans escape of CO₂ is negligible either way, and how much a field protects an atmosphere is debated.

Problem 4 Mercury and Mars: two ways to end

Mercury and Mars sit at the inner and outer edges of the terrestrial planets. They diverged completely. This problem collects three independent measurements, each marking one step in that divergence.

(a) Mercury's spin period P_{rot} and orbital period P_{orb} are given in the Constants and data box. The length of a solar day, from noon to noon, satisfies $1/T_{\text{solar}} = 1/P_{\text{rot}} - 1/P_{\text{orb}}$. Compute T_{solar} and express it in Mercury years.

Solution.

$$\frac{1}{T_{\text{solar}}} = \frac{1}{58.65} - \frac{1}{87.97} = 1.70503 \times 10^{-2} - 1.13675 \times 10^{-2} = 5.6828 \times 10^{-3} \text{ d}^{-1},$$

so $T_{\text{solar}} = 175.97 \text{ d}$. This is almost exactly two Mercury years ($2 \times 87.97 = 175.94 \text{ d}$). The spin and orbit are locked in a 3 : 2 resonance, three rotations for every two orbits, so a solar day on Mercury lasts two of its years: the Sun rises, and the next sunrise comes two orbits later.

I $T_{\text{solar}} = 175.97 \text{ d}$, which equals almost exactly 2 Mercury years (175.94 d) because Mercury is locked in a 3 : 2 spin-orbit resonance.

(b) As Mercury's interior cooled, its radius shrank by ΔR , given in the Constants and data box. Estimate the fractional decrease in surface area, and connect it to the lobate scarps seen across the surface.

Solution. For a sphere $A = 4\pi R^2$, so $\Delta A/A = 2 \Delta R/R$:

$$\frac{\Delta A}{A} = \frac{2 \times 7}{2440} = 5.7377 \times 10^{-3} \approx 0.57\%.$$

The surface had to shrink by about half a percent, and a solid crust accommodates that only by breaking. The result is a global system of lobate scarps, thrust faults where one block of crust has been pushed up over another. Their near-random orientations show the contraction acted in all directions, as a global radius decrease requires, and their total shortening measures how much the interior has cooled since the crust formed. Mercury has a single, globally contracting lithosphere, in contrast to Earth's segmented plates.

I $\Delta A/A = 5.7377 \times 10^{-3} \approx 0.57\%$: global cooling caused Mercury's surface to contract by about half a percent, producing a global network of thrust-fault lobate scarps.

(c) Mercury, the smaller planet, still has a magnetic field; Mars lost its field within its first 400 to 800 Myr. Mercury has no atmosphere; Mars kept a thin one. In a short paragraph, state what planet size predicts for the cooling rate and the dynamo lifetime, why Mercury's dynamo does not follow the lifetime prediction, and what the two planets show about magnetic fields and atmospheric retention. Use the ideas of cooling rate and escape from this worksheet.

Solution. A small planet has a large surface-area-to-volume ratio, so it cools fast: the diffusive cooling time scales as L^2 , and a body half the size cools four times faster. A dynamo needs a core that keeps losing heat to the mantle, or an inner core that freezes and supplies compositional buoyancy, so size alone predicts that the smaller planet loses its dynamo first. Mars follows this prediction: its stagnant-lid mantle, depleted in radioactive elements, stopped cooling the core efficiently, no inner core has formed, and the dynamo died after 400 to 800 Myr. Mercury breaks it through its composition: its iron core fills about 83% of the radius, most of it has frozen, and the thin liquid shell that remains still convects as it freezes and drives a weak field, about 1% of Earth's. The atmospheres do not follow the fields. Mercury has a field and no atmosphere: it formed close to the Sun with few volatiles, and it is small and hot, so light gases have a low Jeans parameter and leave; only a surface-bounded exosphere remains. Mars has no global field and a thin atmosphere: its gravity holds CO_2 against thermal escape, as Problem 3 showed, while the solar-wind channels of part 3(d) thinned it over billions of years. Size sets the cooling rate and, with composition, the dynamo lifetime; volatile supply, gravity and temperature set what an atmosphere keeps.

Size sets the cooling rate, and with it the dynamo lifetime unless composition intervenes: the core of Mars stopped losing heat and its dynamo died, while the oversized core of Mercury still freezes in a thin liquid shell and keeps a weak field despite the smaller size. Atmospheres follow volatile supply, gravity and temperature rather than the field: Mercury has a field and no atmosphere, Mars has no global field and a thin one.